

Autumn 2026 Data Science Clinic

Clinic Overview

The Data Science Clinic is a project-based course where students work in teams as data scientists with real-world clients under the supervision of instructors. Students are tasked with producing deliverables such as data analysis, research, and software along with client presentations and reports. Through the clinic course, Affiliate members gain access to undergraduate or graduate student teams to work on data science projects and explore proof of concepts while identifying top student talent. Projects are tailored and scoped to address company objectives with all deliverables overseen by the Clinic Director.

These unique collaborations allow Affiliate members to supplement their internal data science teams with outside support and perspectives, enlarging their capacity to experiment with new ideas. They also give students a window into a data science career, learning how companies build and use these tools internally.

Clinic Structure

Data Science Clinic runs during Fall, Winter and Spring quarters. Clinic projects are generally scoped to run for two full quarters. Each student works between 10 to 15 hours a week. Each team has a weekly 1-hour meeting with their assigned mentor and must submit a weekly progress report. Mentors are drawn from research staff, postdoctoral fellows and the faculty, subject to availability, interest and needs of the project. The mentor provides intellectual guidance, direct feedback to students and serves as a sounding board for both challenges and direction. The mentors will also provide support and guidance on any gaps in data science knowledge by providing literature and resources. Regular meetings are scheduled as it suits the client needs and to provide feedback to students.

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Accountability Counsel

AI for Human Rights

Background:

Many international finance organizations maintain “accountability mechanisms”—systems for individuals and communities who have observed violations (environmental, labor, human rights) committed by internationally financed companies to submit a formal complaint to the finance organization. This offers communities an avenue to advocate for their rights, but the accountability mechanisms can be complex and difficult to utilize. Accountability Counsel’s mission is to help local communities craft and submit successful complaints against human rights and environmental violations.

Project:

Over the years, Accountability Counsel has collected data on thousands of complaints, including the text of the complaint and whether the complaint was deemed eligible and ultimately successful, and they’re looking to use the data they’ve collected to develop tools to help others write eligible complaints and advocate successfully. This project aims to develop AI models to characterize the likelihood that a given complaint will pass eligibility criteria, given the text of the complaint and relevant context about the accountability mechanism.

Mentor:

Lani Inverarity is Director of Programs & Strategy at Accountability Counsel, where she oversees a team of 13 across the Communities, Policy, and Research programs and leads cross-organizational work on strategy, security, wellness, safeguarding, and evaluation. She joined in 2015 on the Global Communities team, supporting communities harmed by projects financed by international financial institutions and multinational corporations — from farmers displaced in Haiti to herders in Mongolia's Gobi — and has twice served as Acting Executive Director. A former Assistant Crown Counsel in New Zealand and High Court clerk, she holds an LL.M. from Yale Law School and a first-class LL.B. from Victoria University of Wellington.

David Jacobson is the Director of Engineering & Data Science for the Data Science Institute. David specializes in leading machine learning and data science teams to develop practical solutions to complex real-world problems. His work is focused on data science projects for social good, especially related to climate, agriculture, human rights, and global health. Prior to joining DSI, David has led data science teams in biotech, academia, and software technology, most recently at Meliora Therapeutics, an AI startup for small-molecule cancer drug development, and for the Colorado School of Public Health COVID-19 Modeling Group, where his team developed novel approaches to modeling the spread of COVID-19 and the impact of government policy. He holds a BA in Mathematics from Amherst College.

AICE: AI for Climate ⚙️

The added value of added data in weather forecasting

Background:

AI for Climate (AICE) is an interdisciplinary initiative that leverages advances in AI and expanding data availability to tackle critical challenges in climate prediction, impact modeling, and adaptation strategies. By uniting expertise from climate science, computer science, economics, physics, public health, and other fields, AICE develops novel tools—such as physics-informed predictive models and trustworthy datasets—to better understand climate dynamics and inform effective responses.

Project description:

Weather forecasts, provided today as a routine service, are a feat of massive cooperation between direct measurements, physics-based equations for atmospheric physics, and computer systems large enough to solve these equations numerically. Though miraculous, the technology has made weather prediction abstract and poorly understood by the public. The stunning recent success of Artificial Intelligence to forecast weather suggests that data alone is sufficient, without relying on physical models and possibly overcoming their limitations. In this clinic project, you will first build your own predictive models for temperature, precipitation and wind, learning from station-based weather measurements. By systematically increasing the complexity (width and depth) of machine learning models, as well as the geographical breadth and temporal depth of training data, you will uncover the limitations of both and infer the directionality of information propagation in the system, which ultimately traces back to physics. You will then work to answer: where should we build the next weather station? At the same time, you will collect your own measurements with simple sensors you can set up at your own home, and contribute to a growing set of citizen science data.

Mentor:

Justin Finkel is a DSI Postdoctoral scholar and works on advancing the science of extreme weather and its relationship to climate, drawing on classic and new tools from applied mathematics to (1) sample extreme events more efficiently, (2) statistically analyze extreme weather ensembles to infer physical mechanisms and sources of predictability, and (3) inform model development as a result. He obtained a Bachelor's degree in physics and mathematics at Washington University in St. Louis, followed by a PhD in Computational and Applied Mathematics (the first cohort) at the University of Chicago, with an extended visit to the Courant Institute at New York University. He also completed a postdoctoral appointment at MIT.

Apollo Care

Estimation of Deductibles

Background:

Apollo Care is a Chicago-based healthcare technology company that specializes in developing patient access and analytics solutions for pharmaceutical manufacturers, helping make prescription medications more affordable and accessible through innovative technology platforms. The company operates from downtown Chicago and serves pharmaceutical manufacturers as its primary market, offering integrated services including copay programs, hub solutions, and commercial analytics that optimize costs and improve patient outcomes.

Project:

Pharmaceutical companies often give discounts on prescription medicines to patients who meet certain economic or other criteria. One input into these discounts is the deductible, or the amount a patient has to pay before their insurance kicks in. The purpose of this project is to estimate the deductible for patients using medical payment data.

Having accurate deductible estimates for patients will enable pharmaceutical companies to more efficiently use discounting to provide better service to patients. Building an accurate model based off of medical claim data is the first step toward increasing this efficiency.

Mentor:

Austin Nafziger is the Senior Vice President of Product Strategy at Apollo Care, overseeing the firm's innovation roadmap. Over his career, Austin has helped launch several successful startups that have focused on improving efficacy of patient access programs; combatting fraud, waste, and abuse of traditional copay programs; and developing new tools and technologies to help manufacturers create strategic access for their products. He has also contributed to developing and managing pharmacy networks, hub programs, and distribution solutions. Austin holds a Bachelor of Science from the University of Michigan.

BankTrack & Inclusive Development International

S.E.C. Commercial Loan Disclosure Pipeline

Background:

Internationally financed projects like dams, mines, and oil pipelines are notorious for environmental and human rights abuses, including forced displacement of Indigenous people, poisoned water sources, child labor, and physical and sexual abuse by foreign workers. Tracing the investment and supply chains associated with these projects allows researchers and community activists to identify the “pressure points” most responsive to advocacy, such as highly public organizations concerned with their institutional reputations or organizations with prior expressed commitments to accountability and sustainability.

To increase transparency within the finance industry and empower advocates, the DSI, BankTrack, and Inclusive Development International have been working on a suite of tools to map the flow of finances across corporations. One piece currently in development is the Commercial Debt Tracker which pulls mentions of commercial loans in SEC filings and formats them into a public database.

Project Description:

Students will explore the public database performing data analysis and data cleaning, asking important questions about the quality of the data and the state of commercial debt — what’s missing, what results seem incorrect, how does commercial borrowing relate to other macroeconomic trends?. Students will then use modern data visualization techniques to create an interactive dashboard for exploring the data. Lastly, the group will help expand and improve this information extraction pipeline leveraging large language models, traditional machine learning, while including information from new data sources like SEC 6-K forms.

Mentor:

Dustin Roasa: A former investigative journalist for The Guardian, The Los Angeles Times, and The Washington Post, among others—has served as the research director at Inclusive Development International (IDI) since 2016. Roasa’s Follow the Money initiative aims to trace the web of financial transactions and legal relationships surrounding harmful development projects.

Center for Good Food Purchasing

AI Procurement Document Intake for Food Monitoring

Background:

The Center for Good Food Purchasing (CGFP) works with large-scale purchasers to center Good Food values in their operations and decision-making and increase their purchases of Good Food through technical assistance, reporting, relationship-building and peer learning. The Center works with national partners and local grassroots coalitions in cities across the United States to build a cohesive movement in support of Good Food purchasing and to ground Good Food Purchasing Program efforts in community priorities.

Project Description:

The Center reviews receipts and invoices to track the kinds of foods that public institutions purchase. CGFP staff categorize food items and rate them on factors including sustainability, worker treatment, and animal welfare, which institutions can review these ratings when making future purchasing decisions.

Currently, CGFP employees process incoming procurement documents from its partners, which takes large amounts of effort and can introduce human error into the pipeline. The student team will develop AI models to automatically process unstructured procurement documents into a structured format, and will work with CGFP to deploy their models into a production environment. Students will weigh performance versus cost and identify opportunities to incorporate human feedback to ensure high quality model outputs.

Mentor:

Ben Shaman is a Data Analyst at the Center for Good Food Purchasing, where he turns institutional procurement and supply chain data into actionable insights that help large buyers shift toward more sustainable and equitable food systems. His work includes supply chain research assessing alignment with the Good Food Purchasing standards across sustainability, equity, labor, animal welfare, and nutrition, along with reports and visualizations that support institutional decision-making. He came to the role from Dartmouth, where he studied Data Science and Sustainable Food Systems, worked as a research assistant building LLM-based labeling pipelines for large social media datasets, and founded a pop-up restaurant sourcing from local Vermont and New Hampshire farms.

Center for Living System Lab

Cell Biology

Background:

The Center for Living Systems at the University of Chicago brings together researchers from biology, physics, engineering, and data science to study how living systems organize and function. The AI + Quantitative Imaging Shared Facility supports researchers in using computer vision, machine learning, and quantitative image analysis to extract information from biological microscopy data

Project Description:

Time-lapse microscopy allows scientists to observe how individual cells move, divide, and change over time, but manually tracking cells through long image sequences is difficult and time-consuming. Recent advances in AI and deep learning have made automated cell tracking increasingly powerful, although performance can vary across different imaging conditions, biological samples, and image dimensions.

In this project, students will explore and compare state-of-the-art AI and computer vision methods for cell tracking using publicly available annotated datasets from the Cell Tracking Challenge. The team will evaluate how well these methods work on representative 2D and 3D microscopy datasets, with a particular interest in methods that can be applied to 3D live-cell imaging. Promising approaches will then be tested on a real 3D live-cell dataset from a University of Chicago research lab. The goal is to identify practical methods for reliable automated tracking, understand their strengths and limitations, and provide guidance for future biological image analysis.

Mentor:

Liya Ding is a Data Scientist at the University of Chicago Data Science Institute and leads the AI + Quantitative Imaging Shared Facility at the Center for Living Systems. Her work focuses on applying computer vision and deep learning to microscopy, including cell segmentation, tracking, and quantitative analysis of biological images.

Chicago Blackhawks

Corporate Partnerships Lead Generation Agent

Background:

The Chicago Blackhawks started their journey as one of the NHL's "Original Six" professional ice hockey teams in 1926. Members of the Central Division in the NHL's Western Conference and six-time Stanley Cup Champions (1934, 1938, 1961, 2010, 2013, 2015), the team calls the United Center home.

Each day the Team works to uphold our mission to reimagine the potential of hockey. As part of their unwavering commitment to drive impact and purpose in the Chicago community, the organization opened the Blackhawks Ice Center (the team's official practice facility and community space for youth hockey development), continues to make expansions to the Chicago Blackhawks Foundation, and acquired the AHL team the Rockford IceHogs and USHL team Chicago Steel.

Project Description:

The Chicago Blackhawks are looking to create a lead generation agent that will help the Corporate Partnerships team identify relevant targets for sales outreach and optimize their sales pipeline. The ideal outcome of this project is an AI agent that given a research target (e.g. an industry, or a specific company), and background data on current partners, deals, assets and contacts, will perform market research using publicly available data in order to identify the best targets for sales outreach.

Mentor:

Paulino Diaz is the Director of Analytics at The Chicago Blackhawks. He has been working in analytics for 7+ years and previously worked in strategic communications and public relations. Paulino received his Master of Public Policy Analysis from The University of Chicago.

City of Chicago, Dept. of Technology and Innovation

Barriers to Small Business Startup & Expansion

Background:

The Department of Technology and Innovation (DTI) is the City of Chicago's central IT agency. DTI manages the City's core technology infrastructure, digital services, data management, data analytics, and applied data science. Our data team applies data science and advanced analytics, including model development, forecasting, and pattern detection, to strengthen decision-making, improve service delivery, and build more accessible, resident-centered digital tools.

Project Description:

Obtaining a building permit is a critical step for entrepreneurs opening or expanding a business in Chicago, and long or unpredictable processing times can translate directly into higher costs, delayed openings, and lost revenue for small business owners. Today, applicants have little visibility into how long their permit will take once it is submitted.

In this project, students will use the City's public building permit data, combined with zoning and neighborhood characteristics, to predict how long a permit will take to be issued at the time of submission and to identify which permit characteristics are associated with longer processing times. The team will focus on common commercial permit types, such as restaurant buildouts, retail improvements, and renovations, and will apply machine learning methods such as random forests and gradient boosting to predict both expected processing time and the probability that a permit will take more than 30, 60, or 90 days. The team will produce a report on predicting building permit processing times in Chicago along with a dataset of actual and predicted processing times by permit type.

Mentor:

Candice Stauffer, PhD, is a data scientist with extensive expertise in predictive analytics, machine learning, Python, and data visualization. She has nearly a decade of professional experience spanning academia, the private sector, nonprofit organizations, and government. Dr. Stauffer earned her doctorate in astrophysics from Northwestern University, where her research applied machine learning methodologies to model the behavior of astrophysical transients.

She currently serves as the Lead for Data Analytics at the City of Chicago, overseeing the development of data-driven strategies to enhance municipal services and inform decision-making processes. In addition, she is the founding president of Refactor 312, a civic technology nonprofit committed to advancing the use of data and technology to strengthen local communities.

City of Chicago, Dept. of Technology and Innovation

Traffic Crash Hotspots

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Project Description:

Traffic crashes impose enormous human and economic costs on Chicago every year. In this project, students will build a street-segment-level dataset from the City's traffic crash records — covering crashes, vehicles, and people — alongside historical traffic congestion data, and identify hotspots that have experienced particularly high rates of traffic crashes. Because high-traffic locations naturally record more crashes, the team will account for traffic exposure so that the models can surface emerging hotspots that historical crash counts alone would miss, using approaches such as spatial clustering. The team will produce a report on predicting future traffic crash hotspots in Chicago and a street-segment dataset combining historical crash measures with predicted future crash and severe-crash risk.

Mentor:

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City of Chicago, Dept. of Technology and Innovation

311 Service Request

Background:

The Department of Technology and Innovation (DTI) is the City of Chicago's central IT agency. DTI manages the City's core technology infrastructure, digital services, data management, data analytics, and applied data science. Our data team applies data science and advanced analytics, including model development, forecasting, and pattern detection, to strengthen decision-making, improve service delivery, and build more accessible, resident-centered digital tools.

Project Description:

Chicago's 311 system is the front door for non-emergency city services: residents report potholes, broken street lights, graffiti, rodents, missed garbage pickups, tree issues, and dozens of other concerns, and the departments that fulfill these requests have finite crews, equipment, and budgets. Demand is seasonal and uneven — potholes spike after winter freeze-thaw cycles, weeds and yard-waste requests climb in summer — and it varies widely across Chicago's neighborhoods. Anticipating demand ahead of time, rather than reacting after backlogs form, helps the City plan resources and allocate them equitably so that no neighborhood gets left behind.

In this project, students will use a monthly panel of 311 request counts to forecast future request volume from its recent history. The team forecast both city-wide and community-specific demand using both traditional time series modeling and ML methods. Once a successful long-term model has been completed, students will incorporate near-term weather forecasts to predict 311 call volume over the next two weeks, giving departments an operational, short-horizon view of incoming demand alongside the longer-range monthly forecasts, which will be integrated into a live dashboard.

Mentor:

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She currently serves as the Lead for Data Analytics at the City of Chicago, overseeing the development of data-driven strategies to enhance municipal services and inform decision-making processes. In addition, she is the founding president of Refactor 312, a civic technology nonprofit committed to advancing the use of data and technology to strengthen local communities.

Climate Cabinet

Campaign Finance Data

Background:

Climate Cabinet is a nonprofit that provides policy analysis for politicians looking to run on a climate platform or introduce climate legislation. In smaller, local elections, politicians typically do not have the resources to do in-house analysis. Climate Cabinet would like to empower local and state candidates to discuss the political influence of the fossil fuel industry relative to the clean energy industry vis-à-vis campaign financing. In 2010, the U.S. Supreme Court ruled in *Citizens United v. Federal Election Commission* that the First Amendment protected “money as speech” for labor unions, nonprofits, and for-profit corporations. Since then, federal campaign finance contributions from fossil fuel interests have more than doubled—from \$35 million in 2010 to \$84 million in 2018, with most of the funding directed to re-elect candidates with a track record of voting against clean energy policies. However, there is currently no comparable analysis of campaign finance contributions at the state or local level.

Project Description:

DSI has previously built a data pipeline for ingesting state-level campaign finance data from 5 key states for Climate Cabinet. Students will get a chance to explore this real-world, complicated dataset and ask several important questions. They will use data cleaning techniques and record linkage to be able to track entities across state lines. They will use these results to create interactive data visualization to help gain insights into this underexplored and important space.

Mentor:

Grace Rainaldi is a Data Engineer at Climate Cabinet Action, where she has worked since March 2024. She builds the data pipelines and database behind Climate Cabinet's work — connecting local elections, climate champions, and environmental impacts to help climate leaders run, win, and legislate. Her background combines a BS in Earth Systems Science and an MS in Computer Science from Stanford, along with prior engineering roles at Oracle and CIBO.

Fermi National Accelerator Laboratory ⚙

Optimizing a Graph Neural Network (GNN) for Neutrino Event Reconstruction

Background:

Fermilab (Fermi National Accelerator Laboratory) is the United States' premier national laboratory for particle physics and accelerator capabilities. Its mission is to solve the mysteries of matter, energy, space, and time by conducting world-leading research in high-energy physics. The laboratory designs, builds, and operates flagship particle accelerators and detectors, advancing international scientific discovery while pioneering breakthroughs in quantum technology and computing.

Project Description:

The purpose of this project is to improve the way that neutrino detectors measure neutrino interactions. Currently, the particles produced when a neutrino interacts with matter are reconstructed as 3D tracks in a Liquid Argon Time Projection Chamber (LArTPC); the goal of this work is to add more information from other subdetectors such as Photomultiplier Tubes (PMTs) to tighten the overall measurement with additional features. Students will contribute by modifying Python code that implements a Graph Neural Network (GNN), which is a generalization of 2D image recognition to detectors with arbitrary interrelationships. The work will be performed using a large sample of data released by the MicroBooNE experiment, and students will need to use the DSI computing cluster to launch jobs running on many GPUs. Key actions include hyperparameter tuning, evaluating different network configurations to improve information flow between the LArTPC and the other detectors, and assessing performance at local (node) and global (graph) levels. If successful, the results of this study may be published, and contributing students would be co-authors of such a publication.

Mentor:

Giuseppe Cerati is a Scientist in the Fermilab Scientific Computing Division. He is primarily active in liquid argon TPC experiments, such as MicroBooNE, ICARUS/SBN, and DUNE. In these experiments he led searches for anomalous electron neutrino production and measurements of neutrino-nucleus interaction cross-sections. He led several reconstruction-related efforts in the collaborations and is co-author of the NuGraph GNN. Before joining Fermilab he was active in CMS, with highlights of actively participating in the Higgs discovery analysis and leading the Tracking group. He received his PhD in Physics and Astronomy from the University of Milan -- Bicocca in 2008.

Halt the Harm

S.E.C. Commercial Loan Disclosure Pipeline

Background:

Halt the Harm Network (HHN) is an advocacy network that connects leaders, experts, and supporters in the movement to address the harms of pollution. Rather than running its own campaigns, lobbying, or pressuring the industry directly, HHN provides the communication tools, vetted network, expert services, and resources that community activists, researchers, organizers, attorneys, and impacted residents need to win their own campaigns and protect their communities. HHN also provides services on community readiness for advanced ("chemical") plastics recycling threats and on using other tools to reveal oil and gas chemical information.

Project Description:

In this project, students will develop an information extraction pipeline to help Halt the Harm analyze and extract data from news articles and press releases from disparate sources. For these articles to be useful to their activist network, we need to identify what locations or facilities are mentioned in the articles to enable people to discover campaigns happening in their area. Halt the Harm is also interested in other information that can be structured from the articles, such as operating status of the facility mentioned, the type of action announced, and more detailed geographic information. Students will use techniques like entity resolution to determine when multiple articles refer to the same facility.

Mentor:

Veena Singla, PhD is a Senior Fellow with the Halt the Harm Network, where she helps communities prepare to confront proposed "advanced" or chemical plastics recycling facilities. She authored the report "Recycling Lies," which argues chemical recycling is greenwashed incineration, and leads Halt the Harm webinars and briefings that debunk industry claims about the technology. She is also an affiliate of UCSF's Program on Reproductive Health and the Environment and an adjunct assistant professor at Columbia, with a PhD in cell biology from UCSF and prior roles at NRDC and PRHE.

Invenergy

Transformer Data Model & Performance Monitoring

Background:

Invenergy is an American-based multinational power generation development and operations company. The company develops, builds, owns and operates power generation and energy storage projects in the Americas, Europe and Asia, including wind, solar, and natural gas power generation and energy storage facilities. It is North America's largest privately held renewable power generation company and gets millions of live data reads from wind, solar, and other energy source infrastructure. Variations in hardware and procedure make it challenging to do effective performance modeling on the full range of systems for which Invenergy collects data.

Project Description:

This project offers students hands-on experience at the intersection of data science and renewable energy development. Students will develop an asset model for tracking and interpreting data from Invenergy's transformer fleet. Once an effective and consistent data model has been established, students will develop visualizations and predictive models to characterize and monitor the performance of Invenergies hardware fleet. The project will emphasize clear communication of findings through dashboards and visualizations, giving the team the opportunity to deliver actionable insights that could directly influence the future of renewable energy deployment.

Mentor:

Sophie Logan is a Senior Data Scientist on the Enterprise Analytics and AI team at Invenergy, a leading renewable energy developer. Before joining Invenergy, Sophie worked at the World Bank as a climate finance specialist, developing models of countries' environmental metrics, and founded a data science consulting company serving climate technology sectors from building decarbonization to sustainable fashion. She holds a BA in Economics and German from Bard College and an MS in Computer Science and Public Policy from the University of Chicago.

Sophie is passionate about harnessing data and AI to tackle complex challenges in the energy transition and excited to mentor the next generation of data scientists eager to drive innovation at the intersection of technology and climate solutions.

Karczmar Lab, Department of Radiology, University of Chicago ⚙️

Predicting Breast Cancer Treatment Response Using Blood Vessel Networks

Background:

The Karczmar Lab in the University of Chicago Department of Radiology is a leading research group specializing in advanced MRI techniques for cancer imaging and treatment prediction. Led by Dr. Gregory Karczmar, Director of MRI Research, the lab has pioneered innovative MRI methods that are now used in clinical breast cancer screening. The lab combines expertise in MRI physics, medical imaging analysis, and computational methods to develop novel approaches that directly improve cancer detection and patient care.

Project Description:

This project aims to develop a novel approach to predict how breast cancer patients will respond to treatment by analyzing the patterns of blood vessels around tumors. Blood vessels form natural networks that change in response to treatment, and we hypothesize that these vascular patterns can serve as early markers of treatment effectiveness. The ultimate goal is to enable personalized treatment decisions and potentially develop risk assessment tools for earlier cancer detection.

The core innovation involves representing each patient's blood vessel structure as a mathematical graph, where vessel branch points become nodes and vessel segments become edges, annotated with properties like length, width, and curvature. Students will implement Graph Neural Networks (GNNs) to analyze these vascular networks, as GNNs can capture both local structural changes and global network patterns critical for understanding patient-specific responses. The project will involve building a complete pipeline from MRI data to graph representations, training and evaluating GNN models against traditional approaches, and identifying which vascular features are most predictive of treatment outcomes.

Mentor:

Dr. Gregory Karczmar is a Professor of Radiology and Medical Physics and Director of MRI Research, with more than 30 years of experience developing new MRI techniques now used in breast cancer screening. Dr. Milica Medved is a Research Associate Professor of Radiology with over 20 years of experience developing advanced MRI methods for cancer detection and risk prediction. Dr. Zhen Ren is a Research Assistant Professor of Radiology specializing in advanced imaging techniques, including dynamic contrast-enhanced MRI and image reconstruction for breast cancer research.

Pesticide Action Network (PAN)

Forecasting Agricultural Pesticide Use by Chemical Class in California

Background:

The Pesticide Action Network (PAN) North America is an advocacy organization working to end reliance on hazardous pesticides and advance health, resilience, and justice in food and farming. PAN works directly with farmworkers, farmers, and rural communities — those who bear the greatest health and economic burdens of pesticide exposure — and develops data tools that support scientists, advocates, and policymakers seeking to reduce pesticide harm. One of PAN's flagship resources is the California People and Pesticides Explorer, a public-facing platform for viewing, mapping, and downloading California pesticide use data.

Project Description:

To help close this gap, we aim to build a time series forecasting system that predicts monthly pesticide use by active ingredient class, with calibrated uncertainty estimates. The system should allow PAN and allied advocates to generate credible, quantified best-guess estimates of pesticide use for the most recent couple of years, along with confidence intervals that honestly communicate prediction uncertainty. Because different chemical classes have very different seasonal profiles and magnitudes — from fumigants with sharp spring spikes to microbial biologicals with smoother summer peaks — the system should produce separate forecasts per class rather than aggregating across all chemicals. Unlike the pesticide reporting data, weather/climate and economic price signals update with little lag, so these are available as candidate predictors throughout the forecast period — a model can lean on them to project use into months where reported pesticide data does not yet exist.

Mentor:

David Jacobson is the Director of Engineering & Data Science for the Data Science Institute. David specializes in leading machine learning and data science teams to develop practical solutions to complex real-world problems. His work is focused on data science projects for social good, especially related to climate, agriculture, human rights, and global health. Prior to joining DSI, David has led data science teams in biotech, academia, and software technology, most recently at Meliora Therapeutics, an AI startup for small-molecule cancer drug development, and for the Colorado School of Public Health COVID-19 Modeling Group, where his team developed novel approaches to modeling the spread of COVID-19 and the impact of government policy. He holds a BA in Mathematics from Amherst College.

Schmidt: AI in Science ⚙️

Mapping Glacier Crevasses and Glacier-Damage Models

Background:

The department of Geophysical Science at the University of Chicago uses advanced artificial intelligence (AI), machine learning (ML), and physics-based simulations to study how ice and glaciers behave. Ice loss from glaciers is a major contributor to global sea-level rise. Surface crevasses on glaciers are fractures that form where local stresses cause brittle failure of glacier ice. These fractures provide visible indicators of ice deformation and contribute to mechanical weakening, hydrofracture, and iceberg calving. Mapping their distribution is therefore important for identifying regions that may be susceptible to instability and for improving how fracture and damage are represented in glacier models.

Project Description:

This project will use satellite imagery to produce manually traced crevasse maps that will be regarded as expert reference, evaluate automated crevasse-detection algorithms, and test whether continuum damage models reproduce the observed spatial distribution of crevassing. The Clinic team will develop an expert-labeled reference dataset by manually tracing surface crevasses on Pine Island Glacier, a fast-flowing glacier of the West Antarctic Ice Sheet and developing a peer-review process for label uncertainty quantification.

Students will then apply existing automated crevasse-detection algorithms to the same satellite images and compare their outputs with the reference crevasse maps. The students will evaluate the algorithms' detection accuracy, such as precision, recall, spatial overlap, and the distance between mapped crevasse traces. The comparison will identify the strengths and limitations of each algorithm and determine which approaches are most suitable for different crevasse patterns and image conditions.

Mentor:

Weijia (Emma) Liu is an Eric and Wendy Schmidt AI in Science Fellow in the Department of Geophysical Science at the University of Chicago. Prior to this, she obtained her Ph.D. in the Geophysics Department at Stanford University in 2025 under the supervision of Prof. Jenny Suckale, where she studied shear localizations in glaciers using high-performance computing and ML.

Society for the Protection of Underground Networks

Fungal Diversity in Estonia

Background:

The Society for the Protection of Underground Networks (SPUN) is a science-based initiative to map and protect the mycorrhizal networks that regulate the Earth's climate and ecosystems. SPUN's priority is mapping, protecting, and harnessing mycorrhizal fungi. SPUN is activating mycologists around the world, working with local collaborators to obtain core soil samples from all types of ecosystems. The goal is to survey, locate, and identify the diverse domains of mycelia and create a fungal atlas, a biological reference guide to ensure their protection. A key open question for SPUN is whether newer, higher-resolution remote sensing data can produce reliable local biodiversity maps and identify the most valuable places to sample next — DSI has been collaborating with SPUN on this question using Google DeepMind's AlphaEarth foundation model, an environmental embedding of 64 bands on a 10m grid.

Project Description:

A recent regional-scale survey (Tedersoo et al. 2020) measured soil fungal diversity at over 1,200 plots across Estonia — forests, bogs, grasslands, and croplands — using long-read DNA sequencing, making Estonia one of the most densely sampled fungal landscapes on Earth. Alongside richness values, the dataset includes rich plot-level metadata such as soil pH, calcium, vegetation age, and land use. Students will build models that combine AlphaEarth's satellite-derived environmental embeddings with the survey's metadata features to predict the fungal richness measured at each plot. The team will quantify how much predictive power comes from remote sensing versus ground-based measurements, how model quality depends on the number of samples, and produce high-resolution fungal diversity maps of Estonia. The results will directly inform SPUN's global mission: if satellite embeddings plus limited field metadata can reproduce ground-truth diversity, SPUN can extend its maps to finer scales worldwide and target its costly soil sampling where it matters most for conservation.

Mentor:

David Jacobson is the Director of Engineering & Data Science for the Data Science Institute. David specializes in leading machine learning and data science teams to develop practical solutions to complex real-world problems. His work is focused on data science projects for social good, especially related to climate, agriculture, human rights, and global health. Prior to joining DSI, David has led data science teams in biotech, academia, and software technology, most recently at Meliora Therapeutics, an AI startup for small-molecule cancer drug development, and for the Colorado School of Public Health COVID-19 Modeling Group, where his team developed novel approaches to modeling the spread of COVID-19 and the impact of government policy. He holds a BA in Mathematics from Amherst College.

University of Chicago Library

Optimizing Library Storage

Background:

In support of free inquiry and expression, the University of Chicago Library is transforming the global knowledge environment to be open, accessible, and equitable. We enable the University of Chicago and our greater community to create a better world through effective information services, a comprehensive connected collection, and a culture of innovation, respect, and partnership.

Project:

Using anonymized circulation data collected since 2014, students will run analysis on the library's collection to create a probability score of items' likelihood of being checked out. This will be alongside the goal of sorting volumes into cluster groupings to verify the logic of the predictive models students create. The model will be used to create a report that explores distinctive patterns of usage and collection strength within selected Library of Congress subclasses, highlighting areas of unusual demand or uniqueness. Based on this analysis of predictive trends, the Library hopes to make a high quality visual interface so people can interact with clusters and categories of the Library's collection.

Mentor:

David Bottorff is the Collection Management & Circulation Services Librarian for the University of Chicago Library and is leading the Library's strategic priority of developing a comprehensive collection management and storage plan for its physical collections.

University of Northern Iowa

Able Bodies and Minds: Representations of Disability in Classic Young Adult Novels

Background:

The University of Northern Iowa (UNI) is a public university that offers more than 90 majors with a total enrollment of about 9000 students. UNI's main reputation is for its rich history in teacher education.

Project Description:

While individuals with cognitive and physical disability have occasionally been depicted in children's books, their victimized image perpetuates misguided messages about their identity. Characters with minor or significant disability have long been given stereotypical or inconsequential roles and rarely been the main hero of the story. Some recurring patterns in fictional YA novels are disability as tragedy, disability as moral lesson for mainstream individuals, miraculous healing from disability, and disability as overachievement. Sometimes, disability is portrayed as sin and divine punishment. The Clinic team will use traditional natural language processing techniques, statistical methods, and frontier LLMs to explore meaningful patterns in word and language usage within classic Young Adult Novels, identify stereotypical patterns in addressing disability, and present their findings via interpretable data visualization.

Mentor:

Taraneh Matloob is an associate professor of children's literature. She teaches multicultural children's literature as well as instructional design doctoral courses. Her scholarly interests are multicultural children's literature and AI storytelling. Taraneh's most recently published picture books are: *Dear New Friend* and *Invisible River Cross*